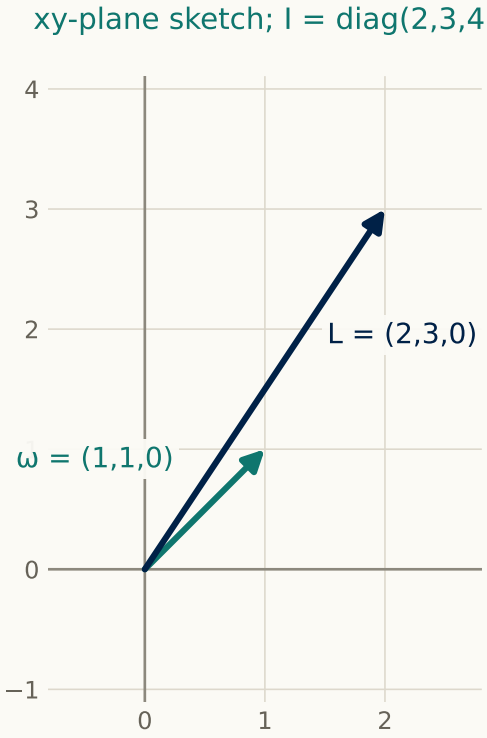Anisotropic inertia can turn angular momentum